Fluid Description

```
    rho_tot += pvecback[pba->index_bg_rho_fld];
    p_tot += w_fld * pvecback[pba->index_bg_rho_fld];
    dp_dloga += (a*dw_over_da-3*(1+w_fld)*w_fld)*pvecback[pba->index_bg_rho_fl
  }
```

```
switch (pba->fluid_equation_of_state) {
case CLP:
  *w_fld = pba->w0_fld + pba->wa_fld * (1. - a);
  break;
```

```
/** - then, give the corresponding analytic derivative dw/da
    by perturbation equations; we could compute it numerical
    but with a loss of precision; as long as there is a simp
    analytic expression of the derivative of the previous
    function, let's use it! */
switch (pba->fluid_equation_of_state) {
case CLP:
  *dw_over_da_fld = - pba->wa_fld;
  break;
```

```
/** - finally, give the analytic solution of the following integral:
    \f$ \int_{a}^{a0} da 3(1+w_{fld})/a \f$. This is used in only
    one place, in the initial conditions for the background, and
    with a=a_ini. If your w(a) does not lead to a simple analytic
    solution of this integral, no worry: instead of writing
    something here, the best would then be to leave it equal to
    zero, and then in background_initial_conditions() you should
    implement a numerical calculation of this integral only for
    a=a_ini, using for instance Romberg integration. It should be
    fast, simple, and accurate enough. */
switch (pba->fluid_equation_of_state) {
case CLP:
  *integral_fld = 3.*((1.+pba->w0_fld+pba->wa_fld)*log(1./a) + pba->wa_fld*(a-1.));
  break;
```